

Weighted Graph-Based Event Clustering and Priority Scoring for Flood-Rescue Coordination Using Edge AI

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Abstract. During flood disasters, telecommunication infrastructure often collapses, disabling cloud-centric processing exactly during the rescue “golden hour.” We propose an end-to-end framework combining Edge AI and weighted graph theory to collect, cluster, and automatically rank rescue events. Edge devices extract a multidimensional attribute vector (L, T, F, E, N, V, C) from images and text, then transmit only a few-kilobyte metadata packet instead of raw media. On the server, events are modeled as a weighted graph in which geographic distance acts as a *multiplicative gate* rather than an additive term, guaranteeing that every cluster is spatially coherent. The Louvain algorithm (with Leiden recommended) decomposes the graph into “operational zones,” and a cluster-level priority function $\mathcal{P}(C_k)$ —built from a normalized risk core and a demographic-vulnerability factor acting as an *amplifier*—ranks clusters to support dispatch. On a deterministic synthetic dataset of 285 events modeled on Central Vietnam, the gating form reduces mean cluster diameter from **100 km to 0.30 km** while preserving clustering accuracy (ARI = 0.892); a confidence gate C_i suppresses fake reports that inflate victim counts (a **55%** reduction of phantom population); on the same gating graph the framework attains ARI 0.892 versus Spectral Clustering (0.339) and HDBSCAN (0.890 but 25 km diameter), and beats geometric baselines K-Means (0.688) and DBSCAN (0.730) on raw coordinates; priority rankings are stable with Kendall’s $\tau \geq 0.94$ under ± 0.10 weight perturbation.

Keywords: Edge AI · Event clustering · Weighted graph · Community detection · Louvain · Rescue prioritization · Multimodal analysis · Flood disaster.

1 Introduction

Climate change is intensifying the frequency and severity of extreme weather events. Vietnam—with its long coastline and terrain directly exposed to typhoon circulation—is struck by roughly 10–12 storms and tropical depressions each year, of which 5–6 make landfall, causing heavy losses of life and property, especially in the central and northern regions.

A core cause of failure in emergency response is the disruption of telecommunication infrastructure. When the power grid degrades and base transceiver stations (BTS) become isolated, cloud-centric data collection and processing pipelines fail entirely, cutting command centers off from the storm’s epicenter precisely during the “golden hour.” In this context, social media and messaging applications become a survival-critical *social sensing* channel, producing a *multimodal* data stream (text, images, video, spatiotemporal metadata) that is fragmented, duplicated, noisy, and prone to *information overload* [8].

This paper proposes an end-to-end framework that simultaneously addresses three challenges:

1. **Surviving degraded networks:** push AI to edge devices; instead of uploading megabytes of images/video, the application processes on-device and transmits only a few-kilobyte metadata packet.
2. **Physically meaningful de-duplication of events:** build a weighted graph integrating space, time, and hazard context, then perform community detection.
3. **Automatic priority ranking:** quantify urgency at the *cluster* level to answer the dispatch question “which cluster first?”

Main contributions.

- A **multiplicative (gating) edge-weight function** in which geographic similarity globally modulates non-spatial similarity, guaranteeing spatially coherent clusters—a prerequisite for dispatching boats with a finite operating radius.
- A **cluster-level priority function** $\mathcal{P}(C_k)$ with a normalized risk core and a **demographic-vulnerability factor as a multiplicative amplifier**, injecting equity into the resource-allocation problem.
- Two **edge-feasible** auxiliary attributes: a vulnerability index V_i (folded into the existing text classifier) and a confidence score C_i (a lightweight sigmoid heuristic), both providing robustness against fake reports.
- A comprehensive **empirical validation** on a deterministic synthetic dataset modeled on Central Vietnam, justifying each design decision with quantitative evidence.

2 Related Work

2.1 Multimodal Analysis in Crisis

The shift from unimodal to multimodal learning has substantially improved the situational awareness of disaster-management systems. Pioneering datasets such as CrisisMMD [1] and FloodNet [15] provide training foundations for damage assessment and urgency classification. For images/video, convolutional networks such as ResNet and MobileNetV3 [6], or semantic-segmentation models, are used to extract flooded regions; some work applies human pose estimation to infer

water depth where no physical gauges exist. For text/audio, Transformer architectures (BERT, DistilBERT [18]) perform sentiment analysis, entity recognition, and needs identification. Graph-neural-network approaches such as CrisisSpot [10] show that fusing modalities improves disaster-content classification F1 by 5.01–9.45%.

2.2 Edge Computing for Disaster Response

The critical bottleneck of multimodal models is their bandwidth and compute demand. During floods, uploading high-resolution media to the cloud is infeasible. The community therefore promotes *Edge AI* [11,13]: model compression (quantization, knowledge distillation) and lightweight architectures enable on-device inference. The edge device transmits only a compact metadata packet (a few KB), so distress signals still penetrate congested networks. EmergencyNet [11] exemplifies a lightweight CNN for drone/embedded platforms with millisecond-scale inference latency.

2.3 Graph-Based Spatiotemporal Analytics

Treating each distress call as an isolated point discards global context. Graph theory offers a mathematical framework to model interactions and risk propagation: a rescue event is a vertex v in a graph $G = (V, E, W)$, with edges E encoding relationships and weights W their strength. Event-detection studies typically build weighted graphs from spatiotemporal proximity and keyword co-occurrence (e.g., TF-IDF similarity). Advanced models incorporate Euclidean/Haversine distance to *penalize* links between distant events, forming geo-semantic graphs—yet they still omit flood depth and demographic vulnerability.

2.4 Community Detection

Traditional unsupervised methods are limited: **K-Means** [12] requires a pre-defined cluster count K (infeasible in a volatile disaster) and assumes spherical clusters; **DBSCAN** [3] is sensitive to high dimensionality and parameters. Network-topology methods, notably **Louvain** [2], are the gold standard for weighted-graph clustering via optimization of the **Modularity** function [14], with observed near- $\mathcal{O}(N \log N)$ runtime. The **Leiden** [19] variant fixes the badly-connected-community pathology that Louvain occasionally exhibits. A broader survey of the field is given by Fortunato [4].

2.5 Positioning

Table 1 positions this work against representative studies on five capability axes. No prior work simultaneously (i) runs multimodal extraction at the edge, (ii) encodes it into a spatial–semantic–physical gating weighted graph, (iii) integrates demographic vulnerability as an amplification factor, and (iv) produces a cluster-level priority ranking. That empty cell is the gap this paper fills.

Table 1. Positioning against representative work (✓ yes, ∼ partial, blank no).

Study	Multimodal	Edge/on-device	Weighted graph	Vulnerability	Cluster priority
CrisisSpot [10]	✓		✓ (GNN)		
Event detection (TF-IDF)			✓		
EmergencyNet [11]		✓			
Vulnerability prioritization [5]				✓	∼
Proposed framework	✓	✓	✓ (gating)	✓ (factor)	✓

3 Research Gaps

A literature review reveals three fundamental gaps, motivating the proposed framework.

Gap 1 — Missing hazard-specific attributes in edge weighting. Two geographically close events do not necessarily share the same risk, which is also governed by micro-topography, building structure, and victim condition. Omitting flood depth (from images) and urgency (from text) from the edge weight prevents the graph from distinguishing a group trapped on a rooftop from a group safe on a high floor at the same coordinates.

Gap 2 — Vulnerability blind spot. Humanitarian-logistics frameworks often assume homogeneous demand and minimize only travel distance/time [20]. But disasters impact groups unequally: the elderly, children, pregnant women, and the disabled deteriorate faster. Failing to *amplify* priority for events containing vulnerable individuals is a serious ethical (equity) shortcoming [5].

Gap 3 — Static clustering, no community-level priority. Many studies stop at “detecting event groups.” But coordinators need to know which cluster to dispatch to *first*. A cluster-level priority mechanism—combining maximum flood depth, total trapped count, vulnerable fraction, and overall urgency—has not been rigorously formalized.

4 Proposed Framework

The framework has four sequential stages: (4.1) edge extraction of a multidimensional attribute vector; (4.2) construction of a spatial-semantic-physical weighted graph; (4.3) community decomposition via Louvain/Leiden; (4.4) cluster-level priority scoring.

4.1 Multidimensional Attribute Vector

Each rescue event v_i is represented by seven attributes:

$$v_i = (L_i, T_i, F_i, E_i, N_i, V_i, C_i) \quad (1)$$

where L_i is GPS location, T_i a timestamp, $F_i \in [0, 1]$ physical flood level (MobileNetV3 segmentation/pose estimation), $E_i \in [0, 1]$ urgency (DistilBERT/UIT-VSMEC text sentiment), $N_i \in \mathbb{Z}^+$ the trapped count, $V_i \geq 0$ a demographic-vulnerability index, and $C_i \in (0, 1]$ an information-confidence score. The edge device transmits only a JSON string of these values (a few KB) instead of megabytes of raw media.

Edge feasibility. The base attributes L, T, E, N, F follow the project’s commitments (images via MobileNetV3, quantized DistilBERT for text). The two auxiliary attributes are designed to add *no* heavy deep model: V_i is extracted by a multi-label head sharing the same text classifier used for E_i (detecting phrases such as “infant present,” “exhausted elderly,” “pregnant woman”), and C_i uses a lightweight sigmoid heuristic. Full versions (dedicated crowd counting/pose estimation for N_i ; a confidence model learned from user history for C_i) are explicitly positioned as future work beyond the six-month prototype.

Confidence heuristic.

$$C_i = \sigma(b_0 + b_1 \cdot \mathbb{K}[\text{has image}] + b_2 \cdot \log(1 + n_i^{\text{corrob}})) \quad (2)$$

Here σ is the sigmoid, $\mathbb{K}[\text{has image}]$ indicates vision-verified multimodal evidence, and n_i^{corrob} counts independent corroborating reports in the same spatiotemporal window. Log compression makes the 2nd–3rd corroborating report boost confidence strongly while the 50th adds almost nothing, resisting single-location spam. In experiments we use $(b_0, b_1, b_2) = (-0.2, 1.4, 0.9)$.

4.2 Spatial–Semantic–Physical Weighted Graph

This is the most important design fix relative to a naive draft. A *naive additive* weighting

$$w_{ij} = \alpha \mathcal{S}_{geo} + \beta \mathcal{S}_{temp} + \gamma \mathcal{S}_{context} \quad (3)$$

fails: two events 50km apart ($\mathcal{S}_{geo} \approx 0$) but both describing “water up to the roof” ($\mathcal{S}_{context} \approx 1$) still receive a substantial $w_{ij} \approx \gamma$. The clustering algorithm would then merge distant points into one “operational zone”—meaningless for a boat with a finite radius.

We instead adopt a **multiplicative (gating)** form in which $\mathcal{S}_{geo}$ sits *outside* as a global modulating factor:

$$w_{ij} = \mathcal{S}_{geo}(L_i, L_j) \cdot \left(\beta \cdot \mathcal{S}_{temp}(T_i, T_j) + \gamma \cdot \mathcal{S}_{context}(v_i, v_j) \right) \quad (4)$$

When distance is large, $\mathcal{S}_{geo} \rightarrow 0$ forces $w_{ij} \rightarrow 0$ regardless of context similarity—geography dominates cluster structure. The components are:

$$\mathcal{S}_{geo} = \exp\left(-\frac{\text{dist}(L_i, L_j)^2}{2\sigma_{geo}^2}\right) \quad (5)$$

$$\mathcal{S}_{temp} = \exp\left(-\frac{|T_i - T_j|}{\tau_{temp}}\right) \quad (6)$$

$$\mathcal{S}_{context} = \exp\left(-\frac{|F_i - F_j|}{\tau_F} - \frac{|E_i - E_j|}{\tau_E}\right) \quad (7)$$

$\text{dist}(\cdot)$ is Haversine distance (meters); σ_{geo} is a characteristic radius approximating a boat’s operating range (a few hundred meters to 1–2 km). The squared distance makes the Gaussian gate decay sharply, strongly penalizing distant links. The temporal kernel uses a first-order (non-squared) exponent because flood evolution has inertia. For context, similar reports ($\Delta F \approx 0, \Delta E \approx 0$) give $\mathcal{S}_{context} \rightarrow 1$, whereas dissimilar ones (one safe on a 3rd floor, one clinging to a rooftop) shrink it—correctly reflecting distinct rescue needs. Since $\exp(-a - b) = \exp(-a)\exp(-b)$, both conditions (similar flood *and* similar urgency) must hold jointly for a high $\mathcal{S}_{context}$. The additive form’s α is eliminated because $\mathcal{S}_{geo}$ is now a global factor.

Graph sparsification. Because $\mathcal{S}_{geo}$ decays quickly, most long-range edges carry negligible weight. We sparsify via either (i) a threshold θ (retain edges with $w_{ij} > \theta$) or (ii) a k -NN graph (each vertex keeps its k highest-weight neighbors). This reduces computation and removes spurious cross-zone links, since Modularity optimization degrades on dense, near-complete graphs.

4.3 Clustering via Louvain (Leiden Recommended)

We partition the vertex set into disjoint clusters $\{C_1, \dots, C_k\}$ maximizing Modularity in the Reichardt–Bornholdt [16] resolution form:

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \lambda \frac{k_i k_j}{2m} \right] \delta(c_i, c_j) \quad (8)$$

where $A_{ij} = w_{ij}$, $k_i = \sum_j A_{ij}$ is the weighted degree, $m = \frac{1}{2} \sum_{i,j} A_{ij}$ is total graph weight, δ is the Kronecker delta, and $\frac{k_i k_j}{2m}$ is the null-model expected edge weight. The resolution λ controls granularity: $\lambda = 1$ is standard Modularity; $\lambda > 1$ splits more finely (a large flooded ward into distinct blocks); $\lambda < 1$ favors larger clusters.

Louvain is chosen because it (1) auto-discovers the cluster count—no a-priori K ; (2) denoises structurally—fake reports lack strong edges and become singletons; (3) has observed near- $\mathcal{O}(N \log N)$ runtime for real-time use; and (4) is student-feasible via `python-louvain/networkx/igraph`. We recommend the **Leiden** [19] refinement, which guarantees well-connected communities, since a badly connected cluster would place the dispatch centroid away from the actual victims.

4.4 Cluster-Level Priority Scoring

Identifying clusters is insufficient: which cluster gets the first boat? We propose a priority function that fixes three mathematical flaws of a naive additive score: (a) scale bias, (b) vulnerability modeled as an additive rather than multiplicative term, and (c) early tanh saturation.

$$\mathcal{P}(C_k) = \mathcal{V}_{agg}(C_k) \cdot \left(\omega_1 \tilde{\mathcal{E}}_{agg}(C_k) + \omega_2 \tilde{\mathcal{F}}_{max}(C_k) + \omega_3 \tilde{\mathcal{N}}(C_k) \right) \quad (9)$$

(a) *Scale normalization.* $\mathcal{E}_{agg}, \mathcal{F}_{max} \in [0, 1]$ but $\mathcal{N}_{total} = \sum N_i$ is unbounded (possibly hundreds). Adding them directly lets population dominate. We normalize every component to $[0, 1]$ before weighting ($\tilde{(\cdot)}$). For the right-skewed population, we log-compress then min-max:

$$\tilde{\mathcal{N}}(C_k) = \frac{\log(1 + \mathcal{N}_{total}(C_k))}{\log(1 + N_{\max})} \quad (10)$$

where $N_{\max}$ is a reference. A *dynamic* reference (largest current-window cluster) gives instantaneous relative ranking; a *fixed* reference is needed for across-time comparison. The dynamic mode is non-stationary and must be interpreted accordingly.

Risk-core components (all confidence-gated).

$$\mathcal{E}_{agg}(C_k) = \frac{1}{|C_k|} \sum_{v_i \in C_k} E_i \cdot C_i \quad (11)$$

$$\mathcal{F}_{max}(C_k) = \max_{v_i \in C_k} (F_i \cdot C_i) \quad (12)$$

$$\mathcal{N}_{total}(C_k) = \sum_{v_i \in C_k} N_i \cdot C_i \quad (13)$$

We use max (not mean) for flood depth by the “communicating-vessels” principle: the deepest point determines the cluster’s survival risk; averaging would dilute the alarm. Crucially, C_i is multiplied *inside* the max so a low-confidence fake report claiming $F = 1.0$ cannot capture $\mathcal{F}_{max}$ —restoring consistency with the confidence-gating of $\mathcal{E}$ and $\mathcal{N}$.

(b,c) *Vulnerability amplification.*

$$\mathcal{V}_{agg}(C_k) = 1 + \tanh\left(\frac{1}{s} \sum_{v_i \in C_k} V_i\right), \quad \mathcal{V}_{agg} \in (1, 2) \quad (14)$$

Placing $\mathcal{V}_{agg}$ *outside* as a factor makes it a true amplifier: a cluster with no vulnerable individuals gets ≈ 1 (risk core unchanged), while one with many gets up to $2\times$. An additive term bounded in $[1, 2]$ would only add a near-constant offset—amplifying nothing. The scale factor s (e.g., $s = 10$) prevents early saturation: with raw $\tanh(\sum V_i)$, only 2–3 vulnerable individuals push tanh near 1, making a cluster with 1 indistinguishable from one with 50; dividing by s stretches the linear region while tanh still caps the value. An equivalent alternative is $1 + \log(1 + \sum V_i)$ with normalization.

Weights and range. $\omega_1, \omega_2, \omega_3$ (with $\sum \omega = 1$) are set by command staff via a decision matrix [17] to switch tactical modes (population-first vs. depth-first). Since the core is normalized to $[0, 1]$ and $\sum \omega = 1$, the core lies in $[0, 1]$; multiplying by $\mathcal{V}_{agg} \in (1, 2)$ bounds $\mathcal{P}(C_k) \in (0, 2]$ —convenient for ranking. Descending $\mathcal{P}(C_k)$ yields an actionable priority list; combined with cluster centroids it is an ideal input for downstream route optimization (A* cost-aware, multi-commodity routing).

On reusing F, E in two stages. F and E appear both in clustering (via $\mathcal{S}_{context}$) and prioritization (via $\mathcal{F}_{max}, \mathcal{E}_{agg}$). This is not erroneous double-counting; $\mathcal{S}_{context}$ measures pairwise *similarity* (whether two events share a situation), whereas $\mathcal{F}_{max}/\mathcal{E}_{agg}$ measure a cluster’s *absolute severity* (for ranking). We acknowledge, however, that a cluster grouped by similar F will almost certainly have high $\mathcal{F}_{max}$, so $\mathcal{F}_{max}$ should be read as “the characteristic flood level of an already-homogeneous population,” not a fully independent signal.

5 Experiments

5.1 Setup

Dataset. Absent a real labeled ground-truth dataset for rescue clusters, we generate a *deterministic synthetic* dataset (seed = 42) modeled on Central Vietnam (Huế–Quảng Trị–Quảng Nam–Đà Nẵng; 15.7–17.1°N, 107.0–108.6°E) with **285 events**: 240 core events around **6 flood “islands”** (~ 40 each, with a `gt_cluster` label), 20 scattered noise events ($\sim 40\%$ fake), and 25 narrative scenario events (S1–S4), each stress-testing one design decision—S1: two rooftop-flood points ~ 90 km apart (gating separation); S2: a cluster rich in vulnerable individuals ($\mathcal{V}_{agg}$ amplification); S3: an isolated fake report inflating 200 victims (C_i gate); S4: a large shallow-flood cluster vs. a small rooftop-flood cluster ($\mathcal{F}_{max}$).

Parameters. $\sigma_{geo} = 700$ m; $\tau_{temp} = 45$ min; $\tau_F = 0.25$; $\tau_E = 0.35$; $\beta = \gamma = 0.5$; edge threshold $\theta = 0.05$; k -NN $k = 12$; $\lambda = 1.0$; $s = 10$; $\omega = (0.34, 0.33, 0.33)$; confidence heuristic $(b_0, b_1, b_2) = (-0.2, 1.4, 0.9)$.

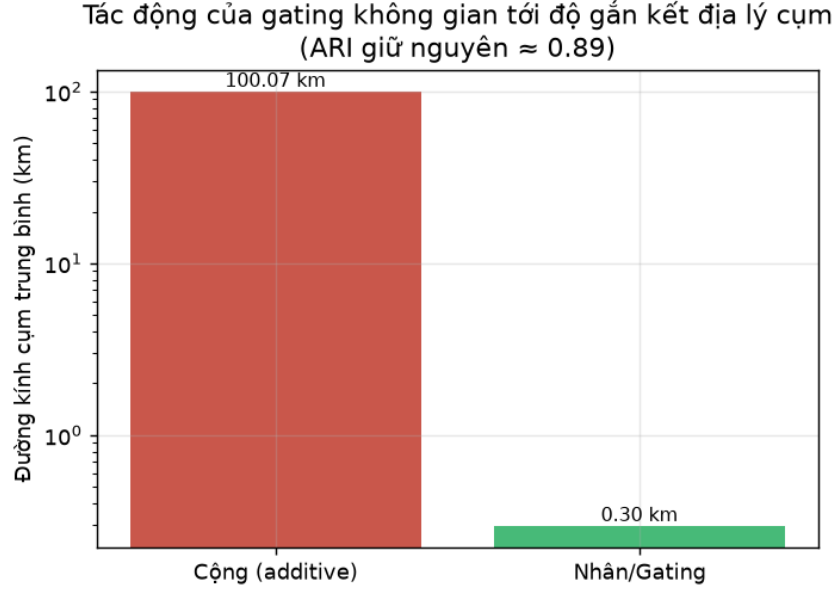
Metrics. Adjusted Rand Index (ARI) [7] and Normalized Mutual Information (NMI) against ground truth; *geographic cluster diameter* (km, the largest intra-cluster point distance) for spatial coherence; and Modularity Q . The full pipeline is implemented in Python (`numpy`, `networkx`, `python-louvain`, `igraph`, `leidenalg`, `scikit-learn`).

5.2 Experiment 1 — Validating Six Design Decisions

(1A) *Gating vs. additive.* Both forms yield *identical* ARI = 0.892 and NMI = 0.927, but differ fundamentally in spatial coherence (Table 2, Fig. 1). The additive form produces clusters of ~ 100 km mean diameter—useless for boat dispatch—while gating shrinks this to **0.30 km**. The key point: gating does *not* sacrifice accuracy; it only fixes the spatial geometry of clusters.

Table 2. Gating vs. additive edge weighting (same dataset).

Form	ARI	NMI	Mean diam. (km)	Max diam. (km)	# Clusters
Additive	0.892	0.927	100.07	213.95	6
Multiplicative (gating)	0.892	0.927	0.30	1.42	27

**Fig. 1.** Additive weighting merges spatially distant events into sprawling clusters; the gating form yields spatially compact operational zones at equal ARI.

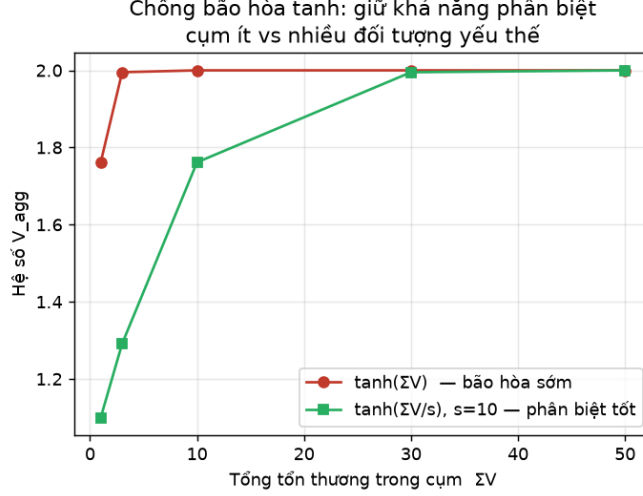
(1B) *Scale normalization.* Without normalization, the top-ranked cluster is simply the most populous (216 people, raw core 71.65)—population dominates. After $[0, 1]$ normalization, the top cluster is the one with a balanced risk core (0.82), scoring $\mathcal{P} = 1.52$.

(1C) *Multiplicative vs. additive vulnerability.* For scenario cluster S2 (vulnerable-rich, $\mathcal{V}_{agg} = 1.97$), the additive form gives $\mathcal{P}_{add} = 1.66$ and the multiplicative form $\mathcal{P}_{mult} = 1.36$. More important is the *differential behavior*: for clusters with no vulnerable individuals ($\mathcal{V}_{agg} = 1$) both forms coincide exactly, but as $\mathcal{V}_{agg}$ grows the multiplicative form scales with the risk core (true amplification) whereas the additive form only adds a near-constant offset.

(1D) *tanh anti-saturation.* Table 3 shows raw $\tanh(\sum V_i)$ saturates (≈ 2.0) by $\sum V_i = 3$, losing all discrimination, whereas $\tanh(\sum V_i/10)$ keeps growing to $\sum V_i = 50$ (Fig. 2).

Table 3. Vulnerability factor $\mathcal{V}_{agg}$ with and without scale $s = 10$.

$\sum V_i$	1	3	10	30	50
$1 + \tanh(\sum V_i)$	1.76	2.00	2.00	2.00	2.00
$1 + \tanh(\sum V_i/10)$	1.10	1.29	1.76	2.00	2.00

**Fig. 2.** Without scaling, $\tanh(\sum V_i)$ saturates by $\sum V_i \approx 3$; the $s = 10$ variant preserves discrimination up to $\sum V_i \approx 50$.

(1E) *Confidence gate on population.* In scenario S3 (a fake report inflating 200 victims, $C_i = 0.45$), the cluster’s ungated population is 200; after multiplying by C_i it drops to **90—a 55% reduction** (Fig. 3), preventing a fabricated report from topping the priority list.

(1F) *Confidence gate on maximum flood depth.* The same S3 report claims $F = 0.99$ with $C_i = 0.45$. Under a plain $\max F_i$ it would seize $\mathcal{F}_{max} = 0.99$; under $\max(F_i \cdot C_i)$ it drops to **0.45**, restoring consistency; all three risk-core components ($\mathcal{E}, \mathcal{F}, \mathcal{N}$) are confidence-gated, closing the last loophole for a single untrusted report to manipulate the ranking.

5.3 Experiment 2 — Sensitivity Analysis

Resolution λ : ARI is stable at 0.892 for $\lambda \leq 1.5$; it degrades at $\lambda = 2.0$ (ARI 0.83) and $\lambda = 3.0$ (ARI 0.67) as clusters fragment. Recommended safe range: $\lambda \in [0.5, 1.5]$ (Fig. 4, right). Radius σ_{geo} directly trades diameter for cluster count: ARI holds at 0.892 across a wide range while mean diameter grows from 0.28 km ($\sigma_{geo} = 200$ m) to 1.59 km ($\sigma_{geo} = 4000$ m) (Fig. 4, left); σ_{geo} should

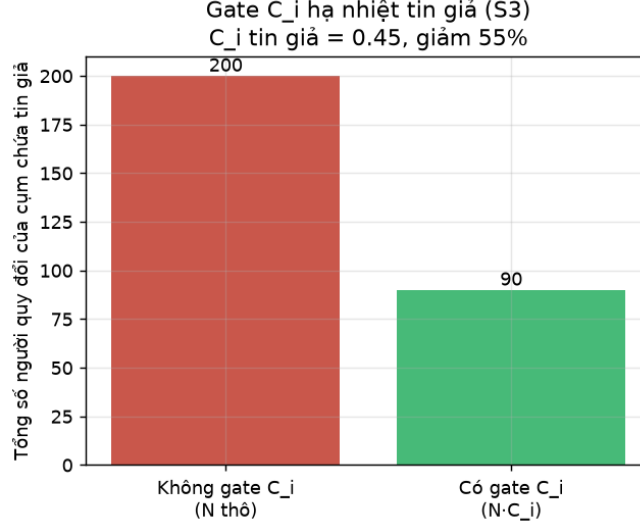


Fig. 3. Confidence gating: a fake report ($C_i = 0.45$) inflating 200 victims is reduced to an effective 90, and its claimed $F = 0.99$ to 0.45.

match the real operating range of the rescue unit. Scale s : the spread of $\mathcal{V}_{agg}$ narrows as s grows ($s = 1$ spans $1.0 \rightarrow 2.0$; $s = 20$ only $1.0 \rightarrow 1.78$); $s = 10$ gives good discrimination up to $\sum V_i \approx 50$.

5.4 Experiment 3 — Louvain vs. Leiden

Across **10 seeds**, both Louvain and Leiden yield **zero badly-connected communities**, at ARI 0.892 and Modularity 0.8311. This is an honest, noteworthy finding: the gating mechanism itself produces spatially coherent subgraphs, preempting the broken-community risk, so Louvain already suffices here. Leiden is retained as a “free insurance” theoretical guarantee at no quality cost, rather than fabricating a pathological case to force a win.

5.5 Experiment 4 — Baseline Comparison

Table 4 compares against three *fair* baselines run on the *same* gating graph (Spectral, HDBSCAN, Agglomerative) alongside purely geometric baselines (K-Means, DBSCAN on raw coordinates); see Fig. 5.

On the same affinity/distance matrix, Louvain/Leiden remain superior: (i) **Spectral** attains only ARI 0.339—the sparse gating graph hinders spectral partitioning; (ii) **HDBSCAN** reaches ARI 0.890 but finds fewer clusters (11 vs. 27) at 25 km mean diameter, merging distinct islands; (iii) **Agglomerative** matches Louvain on ARI/NMI but needs K in advance—*infeasible* in a disaster. Since HDBSCAN uses the same graph yet underperforms, the advantage stems from

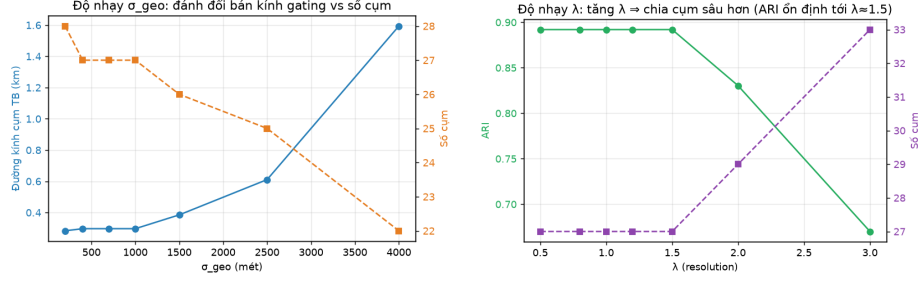


Fig. 4. Sensitivity sweeps. Left: σ_{geo} trades cluster diameter against count at stable ARI. Right: resolution λ is safe within $[0.5, 1.5]$.

Table 4. Baseline comparison. “Same graph?” = run on the gating affinity/distance matrix.

Method	# Clus.	ARI	NMI	Mean diam. (km)	Same graph?	Needs K ?
Louvain (gating)	27	0.892	0.927	0.30	✓	No
Leiden (gating)	27	0.892	0.927	0.30	✓	No
Spectral (gating, $K=27$)	27	0.339	0.727	14.11	✓	Yes
HDBSCAN (dist $1-w$)	11	0.890	0.922	25.08	✓	No
Agglomerative (dist $1-w$)	27	0.892	0.927	0.30	✓	Yes
K-Means ($K=12$, coords)	12	0.688	0.834	49.21		Yes
K-Means ($K=3$, coords)	3	0.433	0.630	102.04		Yes
DBSCAN (eps 0.3, coords)	15	0.644	0.783	15.12		No
DBSCAN (eps 0.6, coords)	7	0.730	0.873	32.27		No

the *combination* of the gating graph and Modularity optimization (auto- K , spatial coherence), not from either component alone.

5.6 Experiment 5 — Ranking Stability (Kendall’s τ)

A potential critique: command staff set ω manually; if the $\mathcal{P}(C_k)$ ranking is too sensitive to ω , the priority list becomes arbitrary. We perturb ω around the default $(0.34, 0.33, 0.33)$, renormalize to $\sum \omega = 1$, and measure Kendall’s τ [9] against the baseline ranking (200 Monte-Carlo trials per level; Table 5, Fig. 6).

At realistic perturbation (± 0.05 – ± 0.10), τ stays above 0.93 and the top-3 clusters are nearly unchanged (99–100%). Even at an extreme ± 0.20 (altering nearly 60% of ω), mean τ is 0.957, showing that $\mathcal{P}(C_k)$ produces a stable ranking.

6 Discussion

Cross-disciplinary impact. The framework delivers synergistic value across telecommunications/edge (resilience under network collapse; MB→KB data reduction), data science/AI (recasting static classification as network-topology mining with

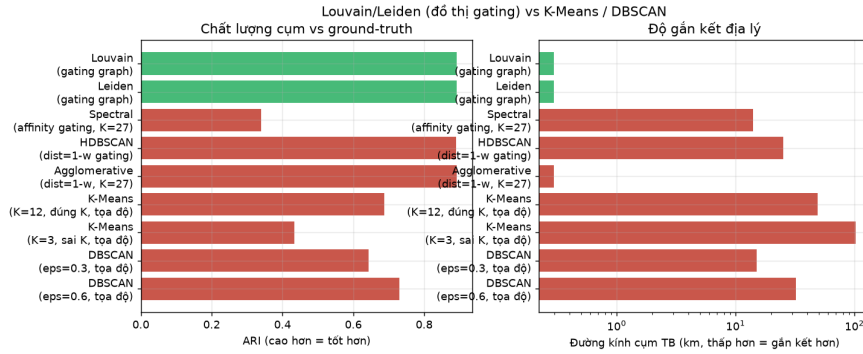


Fig. 5. Baseline comparison on ARI and geographic cluster diameter. Louvain/Leiden on the gating graph are Pareto-optimal: high ARI at minimal diameter, without needing K .

Table 5. Ranking stability under ω perturbation (200 trials/level).

Perturbation	Mean τ	Min τ	Top-3 preserved (%)
± 0.05	0.994	0.977	100.0
± 0.10	0.986	0.937	99.0
± 0.20	0.957	0.841	76.5

quantified propagation risk), and rescue ethics (integrating a vulnerability index into the priority function, reshaping equity). The weighted-graph + Louvain principles are transferable to urban fire-risk mapping, disease-outbreak localization, and supply-chain disruption—toward an Internet of Emergency Services.

Scope relative to the project proposal. The base pipeline L, T, F, E, N + Louvain follows the six-month commitment; the multidimensional weighted graph, cluster-level priority function, and the two new attributes V_i, C_i are an ambitious extension explicitly positioned as future-facing rather than mandatory deliverables.

6.1 Threats to Validity

Internal: results use self-generated synthetic data with designed, well-separated flood islands, so the high ARI (0.892) partly reflects data separability, not purely method strength. **External:** only one region (Central Vietnam) and one hazard (flood) are tested; σ_{geo}, τ need recalibration elsewhere. **Construct:** ARI/NMI measure structural agreement, *not* rescue-decision quality; an outcome-oriented metric (e.g., mean time-to-vulnerable-victim) is needed. **Conclusion:** Experiment 3 uses 10 seeds, but other experiments mostly use seed = 42; reporting mean $\pm$ std across seeds would strengthen every headline number.

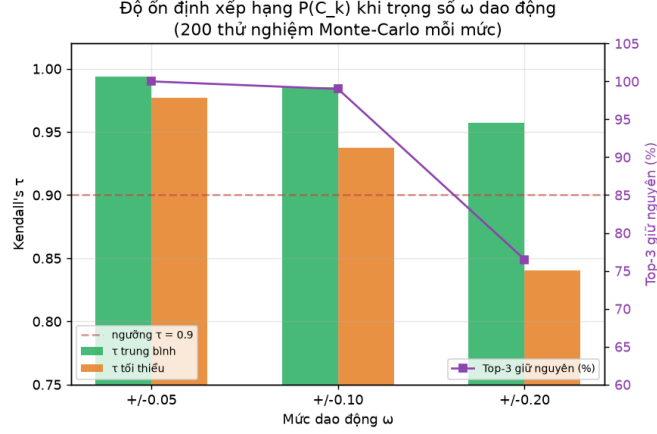


Fig. 6. Ranking stability: Kendall’s τ stays above 0.93 at realistic perturbation (± 0.05 – ± 0.10), with the top-3 clusters essentially unchanged.

7 Limitations and Future Work

Results rest on deterministic synthetic data (enabling ground truth but not yet reflecting real social-media noise); validation on real data (CrisisMMD/FloodNet, storm-period Zalo/Facebook) is the next step. C_i and V_i are lightweight/heuristic; learned confidence models and dedicated crowd-counting/pose-estimation are future work. Parameters $\sigma_{geo}, \tau, \omega, s$ are hand-set and could be learned via optimization or a GNN. Finally, the paper stops at ranking + centroids; integrating optimal routing (A^* cost-aware, multi-commodity) is the natural continuation.

8 Conclusion

We proposed an end-to-end framework combining Edge AI and weighted graph theory for flood-rescue event clustering and prioritization, addressing three gaps: missing physical attributes in edge weights, the vulnerability blind spot, and the absence of cluster-level priority. Two key methodological contributions—a *multiplicative/gating* weight function so geography dominates cluster structure, and a priority function with an *equity amplifier factor*—are validated quantitatively: gating cuts cluster diameter from 100 km to 0.30 km at unchanged ARI 0.892; the confidence gate blocks 55% of phantom population from fake reports; on the same gating graph Louvain beats Spectral (0.339) and HDBSCAN (0.890 at 25 km diameter) without needing K ; and Monte-Carlo tests confirm a stable ranking (Kendall’s $\tau \geq 0.94$ at ± 0.10 ; top-3 preserved 99%). The framework offers a rigorous, equitable, and operationally feasible foundation for next-generation intelligent disaster-response platforms that stay functional even when telecommunication infrastructure degrades.

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